

Multimedia Communications

Sample Exam Answers and Study Notes

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1 Multimedia Representation and Perception

Domanda 1

Explain the **Contrast Sensitivity Function (CSF)**: what it is, in which units **spatial frequency** is measured, where it peaks, and the direct implication for **quantization** in compression.

- **CSF**: HVS sensitivity to luminance contrast as **spatial frequency** changes.
- Unit: **cycles per degree** of visual angle.
- Shape: band-pass → low sensitivity at very low and very high frequencies; peak at intermediate frequencies.
- Compression consequence → frequency-dependent **quantization**:
- visually important frequencies → smaller steps;
- less visible frequencies → coarser steps.

Domanda 2

Compare **cones** and **rods** (number, function, lighting conditions) and explain why the RGB-to-Y conversion weights the green component the most.

- **rods**: high light sensitivity, low-light vision, no color.
- **cones**: color vision, brighter conditions, short/medium/long wavelength classes.
- Luminance example:

$$Y \approx 0.299R + 0.587G + 0.114B$$

- Green coefficient largest → HVS most sensitive to green/yellow luminance detail.

Domanda 3

Explain the J:a:b **chroma subsampling** notation. Compute the data-reduction factor of 4:2:0 vs full RGB and justify why it is perceptually acceptable.

- $J : a : b \rightarrow$ **chroma** samples kept relative to J **luma** samples over two rows.
- 4:2:0 → full-resolution Y , half horizontal and half vertical Cb, Cr .
- Over 2×2 pixels:
- RGB: $4 \times 3 = 12$ component samples;
- YCbCr 4:2:0: $4Y + 1Cb + 1Cr = 6$ component samples.
- Reduction → about 1/2 of full RGB component samples.
- Perceptual reason → eye is more sensitive to luminance detail than chrominance detail.

Domanda 4

Draw the Basic Tools for Compression scheme (Transform -> Prediction -> **quantization** -> **entropy** Coding) and indicate which stage is the only **lossy** one and why.

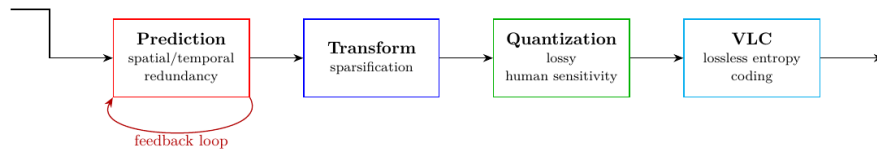
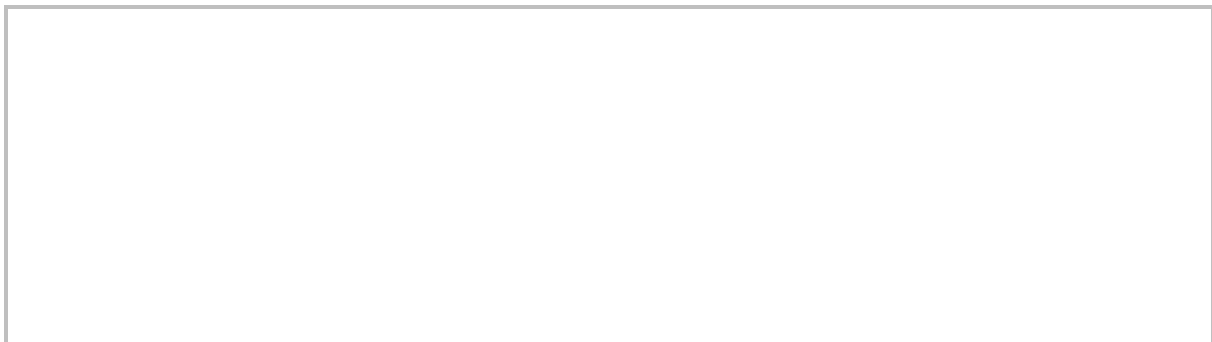


Figure 1: Pasted image 20260624205523



- Prediction / transform → decorrelate signal or compact energy.
- **entropy** coding → **lossless**; shorter codes for more probable symbols.
- **quantization** → only **lossy** stage: many input values map to same reconstruction value → exact original cannot be recovered.

2 Compression, Prediction and Lossless Coding

Domanda 5

Principles of image compression. Discuss the criteria for evaluating a compression algorithm: **rate**, quality, [Bonus: robustness, delay, complexity].

- Compression exploits:
- **statistical redundancy** → correlated samples/blocks/frames;
- **spatial / temporal redundancy** → repeated structures;
- **psychovisual redundancy** → imperceptible **distortion**.
- Types:
- **lossless** → exact reconstruction, lower compression;
- **lossy** → approximate but perceptually close reconstruction, higher compression.
- **rate**:

$$R_{\text{image}} = \frac{B_{\text{out}}}{NM} \quad [\text{bpp}]$$

$$R_{\text{stream}} = \frac{B_{\text{out}}}{T} \quad [\text{bit/s}]$$

- Compression ratio:

$$\text{CR} = \frac{B_{\text{in}}}{B_{\text{out}}}$$

- **distortion** / quality:

$$D(f, \hat{f}) = \frac{1}{NM} \|f - \hat{f}\|^2$$

$$\text{PSNR}(f, \hat{f}) = 10 \log_{10} \left(\frac{V^2}{D(f, \hat{f})} \right)$$

- Perceptual weighting:

$$D_W(f, \hat{f}) = \frac{1}{NM} \|h \star (f - \hat{f})\|^2$$

$$\text{WPSNR}(f, \hat{f}) = 10 \log_{10} \left(\frac{V^2}{D_W(f, \hat{f})} \right)$$

- Tradeoffs → **rate**, quality, robustness, delay, complexity; also **SSIM**, **LPIPS**.

Domanda 6

A zero-mean Gaussian random signal **HAS** an autocorrelation function:

$$r_X(n-m) = E[X(n)X(m)] = \sigma^2 \rho^{|n-m|}$$

> - Consider the predictor $V(n) = X(n-1)$. For which values is the prediction gain positive?
 > - Optimal linear predictor is $V(n) = -\sum_{i=1}^P a_i x_{n-i}$, with $a = -R_X^{-1}r$, $R_X(i,j) = r_X(i-j)$, $r = [r_X(1) \cdots r_X(P)]^T$. Find optimal linear predictor of order $P = 1$ and compute prediction gain, compare then with previous case. > - Compute the optimal predictor of order 2. Compare with the previous cases.

Given:

$$r_X(n-m) = \mathbb{E}[X(n)X(m)] = \sigma^2 \rho^{|n-m|}$$

- Predictor $V(n) = X(n-1)$:

$$\sigma_y^2 = \mathbb{E}[(X(n) - X(n-1))^2] = 2\sigma^2(1-\rho)$$

$$G_P = 10 \log_{10} \left(\frac{1}{2(1-\rho)} \right)$$

- Positive gain iff:

$$\rho > \frac{1}{2}$$

- Optimal linear predictor:

$$V(n) = -\sum_{i=1}^P a_i x(n-i), \quad \vec{a}^{opt} = -R_X^{-1} \vec{r}$$

- For $P = 1$:

$$a_1^{opt} = -\rho \Rightarrow V(n) = \rho X(n-1)$$

$$\sigma_y^2 = \sigma^2(1-\rho^2), \quad G_P = 10 \log_{10} \left(\frac{1}{1-\rho^2} \right)$$

- For $P = 2$:

$$R_X = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad \vec{r} = \sigma^2 \begin{bmatrix} \rho \\ \rho^2 \end{bmatrix}$$

$$\vec{a}^{opt} = \begin{bmatrix} -\rho \\ 0 \end{bmatrix}$$

- Result $\rightarrow$ second tap gives no extra gain for AR(1) source.

Domanda 7

Draw and comment on the schemes of **predictive quantization** (encoder and decoder).

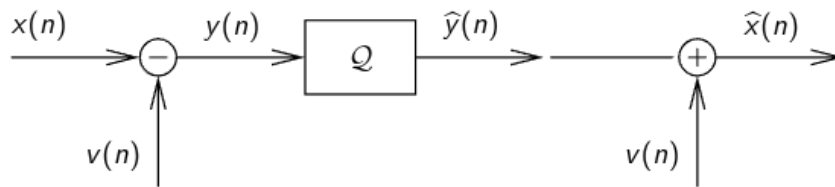
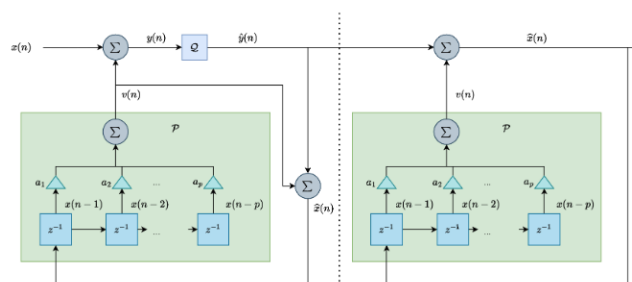


Figure 2: Predictive quantization - open loop

- **Open-loop predictive quantization:**
- subtract predictor $v(n)$ from $x(n) \rightarrow$ residual $y(n)$;
- quantize residual $\rightarrow \hat{y}(n)$;
- add $v(n)$ back $\rightarrow$ reconstruction $\hat{x}(n)$.
- Problem $\rightarrow$ encoder predicts from original samples, decoder from reconstructed samples $\rightarrow$ **drift**.



This is the correct scheme: the predictor is fed with the same data (quantized past samples) both at the encoder and the decoder.

Figure 3: Predictive quantization - correct closed loop

- **Closed-loop encoder** → encoder contains same reconstruction loop as decoder.
- Predictor input matches both sides:

$$y(n) = x(n) - v(n), \quad \hat{x}(n) = \hat{y}(n) + v(n)$$

- Useful iff residual variance is smaller than original variance.

Domanda 8

Discuss the principles of **lossless** coding.

- **lossless coding** → maps symbols to bitstrings; reconstructs original sequence exactly.
- Alphabet/code:

$$\mathcal{X} = \{x_1, \dots, x_M\}, \quad \mathcal{C} : \mathcal{X} \rightarrow \{0, 1\}^*$$

- Fixed-length coding:

$$R = \lceil \log_2 M \rceil$$

- VLC → probable symbols get shorter codewords:

$$\bar{L} = \sum_i p_i l_i$$

- **Prefix code** → no codeword is prefix of another → instantaneous decoding.
- **McMillan theorem** → best uniquely decodable code **HAS** same performance as best prefix code.
- **Kraft inequality**:

$$\sum_i 2^{-l_i} \leq 1 \iff \text{prefix code exists with lengths } \{l_i\}$$

- **entropy** lower bound:

$$H(X) = - \sum_i p_i \log_2 p_i$$

- Shannon:

$$H(X) \leq \bar{L} < H(X) + 1$$

Domanda 9

Consider a source that emits the symbols A, B, C, D, E, and F. The probabilities of these symbols are given in the following table:

$$> p_A = 0.3, p_B = 0.1, p_C = 0.05, p_D = 0.18, p_E = 0.15, p_F = 0.22 >$$

> - Describe the Huffman Algorithm. > - Compute Huffman code for this distribution. > - Compare the average length of the Huffman code with the distribution **entropy**.

- **Huffman algorithm** → repeatedly merge two least probable symbols.
- One optimal code:
- A, $p = 0.30 \rightarrow 10$, length 2
- B, $p = 0.10 \rightarrow 1111$, length 4
- C, $p = 0.05 \rightarrow 1110$, length 4
- D, $p = 0.18 \rightarrow 00$, length 2
- E, $p = 0.15 \rightarrow 110$, length 3
- F, $p = 0.22 \rightarrow 01$, length 2

$$\bar{L} = 0.30 \cdot 2 + 0.10 \cdot 4 + 0.05 \cdot 4 + 0.18 \cdot 2 + 0.15 \cdot 3 + 0.22 \cdot 2 = 2.45$$

$$H(X) = - \sum_i p_i \log_2 p_i \approx 2.406 \text{ bit/symbol}$$

- Overhead $\rightarrow \bar{L} - H(X) \approx 0.044 \text{ bit/symbol}$.

Domanda 10

Why arithmetic encoder is preferred over Huffman for high-performance **lossless** coding?

- Huffman → integer codeword lengths → penalty for non-dyadic probabilities.
- Block Huffman reduces penalty, but alphabet grows as M^K .
- Arithmetic coding → encodes whole sequence as interval in $[0, 1)$.
- **rate:**

$$\mathcal{L} < H(X) + \frac{2}{n} \xrightarrow{n \rightarrow \infty} H(X)$$

- Also suited to adaptive/context probability models.

Domanda 11

(LLCod) Explain the difference between Fixed-Length Coding (FLC) and Variable-Length Coding (VLC), and describe why VLC is theoretically superior for non-equiprobable sources.

- **FLC** → same codeword length for every symbol; simple, instant parsing, inefficient if probabilities non-uniform.
- **VLC** → different lengths; probable symbols shorter.

$$\bar{L} = \sum_i p_i l_i$$

- Superiority condition → non-equiprobable source.

Domanda 12

(LLCod) Discuss the importance of the prefix condition in Variable-Length Coding and how it relates to the concept of instantaneous decodability.

- **Prefix condition** → no codeword is prefix of another.
- Consequence → decoder identifies symbol as soon as codeword ends; no look-ahead.
- Kraft:

$$\sum_i 2^{-l_i} \leq 1$$

- Equality → complete prefix tree.

Domanda 13

(LLCod) What are the two distinct mechanisms by which “block coding” improves the efficiency of **lossless** compression?

1. Sources with memory:

$$H(X^K) \leq \sum_{i=1}^K H(X_i)$$

→ dependencies exploited.

2. Memoryless non-dyadic sources:

$$\frac{H(X^K)}{K} \leq \frac{L^*}{K} < \frac{H(X^K)}{K} + \frac{1}{K}$$

→ integer-length overhead spread over K symbols; overhead/symbol → 0.

Domanda 14

(LLCod) Provide a synthetic comparison between the main **lossless** coding techniques (Exp-Golomb, Huffman, Arithmetic, Dictionary, Neural) in terms of complexity and Latency, and provide a typical use case for each of them.

- **Exp-Golomb**: good for small integers; very low complexity/latency → syntax elements, motion-vector residuals.
- **Huffman**: optimal among symbol prefix codes; low/medium complexity, very low latency → JPEG, DEFLATE-style coding.
- **Arithmetic**: near **entropy**; high complexity, medium latency → **CABAC**, context-adaptive coding.
- **Dictionary (LZ/LZW)**: universal for repeated patterns; medium complexity, low/medium latency → text, GIF, ZIP-like systems.
- **Neural lossless**: potentially very high efficiency; very high complexity/latency → research, high-resolution image models.

Domanda 15

(LLCod) Describe the principle of the Huffman coding. For the following probability distribution, compute the optimal **lossless** code, and compare its average length to the source's **entropy**:

$$> p_A = 0.35, p_B = 0.1, p_C = 0.07, p_D = 0.08, p_E = 0.12, p_F = 0.28 >$$

- One optimal Huffman code:
- A, $p = 0.35 \rightarrow 00$, length 2
- B, $p = 0.10 \rightarrow 100$, length 3
- C, $p = 0.07 \rightarrow 101$, length 3
- D, $p = 0.08 \rightarrow 110$, length 3
- E, $p = 0.12 \rightarrow 111$, length 3
- F, $p = 0.28 \rightarrow 01$, length 2
- Relevant lengths:

$$l_A = l_F = 2, \quad l_B = l_C = l_D = l_E = 3$$

$$\bar{L} = 0.35 \cdot 2 + 0.28 \cdot 2 + (0.10 + 0.07 + 0.08 + 0.12) \cdot 3 = 2.37$$

$$H(X) \approx 2.304 \text{ bit/symbol}$$

- Overhead → $\bar{L} - H(X) \approx 0.066 \text{ bit/symbol}$.

Domanda 16

(LLCod) Which of the following statements best describes the behavior of the Shannon **entropy** $H(X)$ for a binary random variable with probability p ?

For $X \sim \text{Bernoulli}(p)$:

$$H(X) = -p \log_2 p - (1-p) \log_2 (1-p)$$

- $H(X) = 0$ for $p = 0$ or $p = 1$.
- Maximum at $p = \frac{1}{2}$:

$$H\left(\frac{1}{2}\right) = 1 \text{ bit}$$

Domanda 17

(LLCod) Why is Lempel-Ziv (e.g., LZW) considered a “universal” coding algorithm?

- No prior source probability distribution needed.
- Encoder and decoder build same dictionary adaptively from observed sequence.
- Stationary ergodic sources $\rightarrow$ asymptotically optimal.

—

3 Transform Coding and JPEG

Domanda 18

Why the geometric mean of the variances of a random vector is key information to evaluate the **rate-distortion** performance of a **quantizer**?

- High-resolution transform coding with optimal **bit allocation**:

$$D^* = c_{GM} \sigma_{GM}^2 2^{-2\bar{R}}$$

- Orthogonal transforms preserve total energy $\rightarrow$ preserve arithmetic mean:

$$\sigma_{AM,Y}^2 = \sigma_{AM,X}^2$$

- Transform can change variance distribution.
- Good transform $\rightarrow$ few high-energy coefficients, many low-energy coefficients $\rightarrow$ lower σ_{GM}^2 .

$$\sigma_{GM}^2 \leq \sigma_{AM}^2$$

- Lower geometric mean $\rightarrow$ lower **distortion** at same **rate**.

$$G_T = \frac{D_{PCM}}{D_Y} = \frac{\sigma_{AM,Y}^2}{\sigma_{GM,Y}^2}$$

$$G_{T,dB} = 10 \log_{10} G_T$$

Domanda 19

Write the resource allocation problem for transform coding. Derive the Huang-Schulteiss formula.

- High-resolution **distortion**:

$$D = \frac{1}{M} \sum_{k=0}^{M-1} c_k \sigma_k^2 2^{-2R_k}$$

- **rate** constraint:

$$\sum_{k=0}^{M-1} R_k \leq R_{\text{tot}}$$

- Lagrangian:

$$J(\vec{R}, \lambda) = \frac{1}{M} \sum_{k=0}^{M-1} c_k \sigma_k^2 2^{-2R_k} + \lambda \left(\sum_{k=0}^{M-1} R_k - R_{\text{tot}} \right)$$

- Set $\frac{\partial J}{\partial R_k} = 0 \rightarrow$ Huang-Schultheiss:

$$R_k^* = \frac{R_{\text{tot}}}{M} + \frac{1}{2} \log_2 \left(\frac{c_k \sigma_k^2}{c_{GM} \sigma_{GM}^2} \right)$$

$$c_{GM} = \sqrt[M]{\prod_{k=0}^{M-1} c_k}, \quad \sigma_{GM}^2 = \sqrt[M]{\prod_{k=0}^{M-1} \sigma_k^2}$$

- Above geometric mean $\rightarrow$ more bits.
- Below geometric mean $\rightarrow$ fewer bits.

Domanda 20

Explain why the **KLT** is the optimal linear transform for decorrelation and why the **DCT** is used in practice instead.

- **KLT** $\rightarrow$ eigenvectors of signal covariance matrix.
- In **KLT** basis $\rightarrow$ coefficients decorrelated; optimal energy compaction among linear orthogonal transforms for given statistics.

$$T_{KLT} = [u_1 \ u_2 \ \cdots \ u_M]^T$$

- Problem $\rightarrow$ **KLT** depends on actual image/block statistics; encoder must estimate covariance, compute transform, signal it.
- **DCT** used because fixed, separable, fast, no side information, good **KLT** approximation for locally correlated image blocks.

Domanda 21

Describe the principles of the JPEG standard.

- **JPEG** $\rightarrow$ **lossy** still-image codec based on block **DCT**, **quantization**, **entropy** coding.
- Standard mainly defines decodable bitstream and decoder behavior; encoder choices flexible.

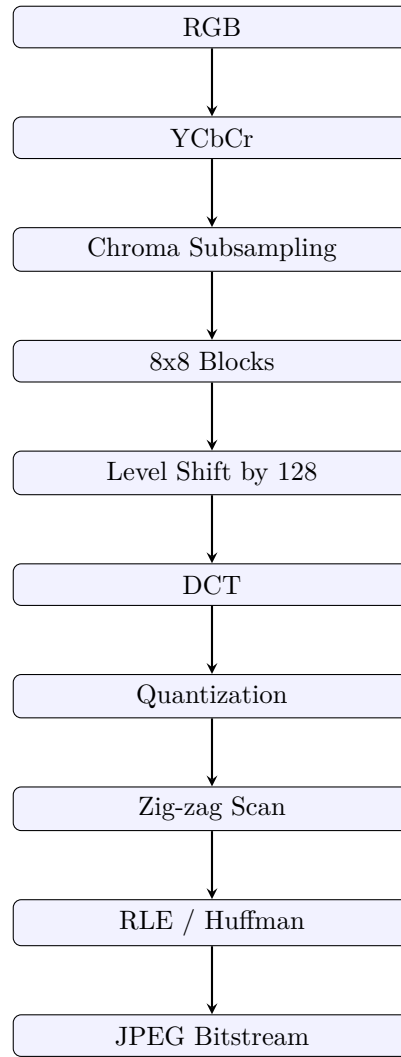


Figure 4: JPEG Baseline Encoding Chain

- **quantization:**

$$\tilde{C}_{ij} = \text{round} \left(\frac{C_{ij}}{q_{ij}} \right)$$

- **quantization** table $\rightarrow$ smaller steps for important low frequencies, larger steps for high frequencies.
- Quality factor:

$$S_F = \begin{cases} \frac{5000}{Q} & 1 \leq Q \leq 50 \\ 200 - 2Q & 50 < Q \leq 99 \\ 1 & Q = 100 \end{cases}$$

$$q \leftarrow \frac{S_F}{100} q^*$$

- Metadata:
- **JFIF** $\rightarrow$ density, resolution, thumbnails;

- **EXIF** → date/time, GPS, camera settings.

Domanda 22

(TC&JPEG) Draw the scheme and describe the functional blocks of a JPEG encoder.

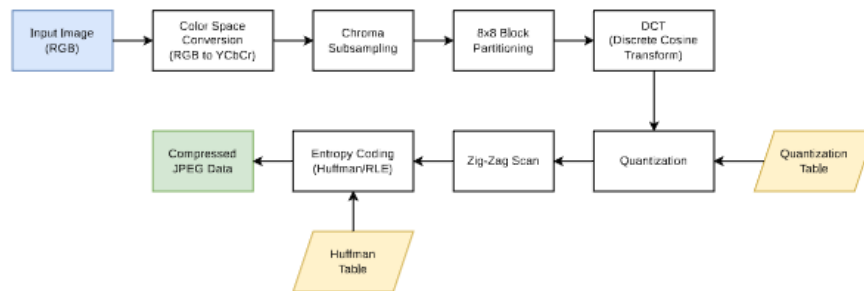


Figure 5: JPEG encoder

- **DCT** → energy compaction.
- **quantization** → **lossy** coefficient reduction.
- Zig-zag scan → groups zeros.
- **entropy** coding → **lossless** compression.

Domanda 23

(TC&JPEG) Describe the problem of “frequency leakage” in the Discrete Fourier Transform (DFT) when applied to signal compression and how the **discrete cosine transform (DCT)** addresses it.

- DFT assumes finite signal is periodic.
- Boundary mismatch → artificial discontinuities → energy spread to high frequencies = **frequency leakage**.
- **DCT** uses symmetric extension → smoother periodic continuation → better energy compaction for images.

Domanda 24

(TC&JPEG) Explain the **entropy** coding process for AC coefficients in the JPEG standard and the significance of the “End of Block” (EOB) symbol.

- After zig-zag scan, AC coefficients $\rightarrow$ pairs:

$$(r, k)$$

- $r \rightarrow$ run of preceding zeros.
- $k \rightarrow$ category of non-zero amplitude.
- Special symbols:

$$(15, 0) = \text{ZRL, run of 16 zeros}$$

$$(0, 0) = \text{EOB, End Of Block}$$

- **EOB** $\rightarrow$ all remaining block coefficients are zero.

Domanda 25

(TC&JPEG) Describe the JPEG **lossless** coding process applied to the following table of quantized **DCT** coefficients:

$$> \begin{bmatrix} > 10 & 3 & 0 & 0 & 0 & 0 & 0 & 0 \\ > -2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ > 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ > 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ > 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ > 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ > 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ > 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 > \end{bmatrix} >$$

- Treat missing positions as zero-padded 8×8 block.
- DC:

$$DC_n = 10$$

- JPEG sends DC difference:

$$\Delta DC = DC_n - DC_{n-1}$$

- First block with $DC_{n-1} = 0 \rightarrow \Delta DC = 10$.
- Non-zero AC in zig-zag order:

3, -2, 1, 1, 4, 1

- Run/category:
- 3 → run 0, category 2
- -2 → run 0, category 2
- 1 → run 0, category 1
- 1 → run 0, category 1
- 4 → run 6, category 3
- 1 → run 1, category 1
- EOB → run 0, category 0
- No ZRL → no run longer than 15 zeros before non-zero coefficient.

Domanda 26

(TC&JPEG) What is the primary purpose of an orthogonal transform in the context of the transform coding paradigm?

- Primary purpose → **sparsification** / **energy compaction**.
- Orthogonal transform preserves energy and squared error:

$$T^{-1} = T^T, \quad \|TX\|^2 = \|X\|^2$$

Domanda 27

(TC&JPEG) In the context of JPEG compression, what is the role of the **quantization** table?

- **quantization table** → **quantization** step for each **DCT** coefficient.
- Controls **rate**-quality tradeoff.
- Reflects visual sensitivity:
- low frequencies → smaller steps;
- high frequencies → larger steps.

Domanda 28

(TC&JPEG) Which statement regarding the relationship between the Arithmetic Mean (AM) and Geometric Mean (GM) of variances in transform coding is correct?

- Orthogonal transforms preserve energy → preserve arithmetic mean of coefficient variances.

- They do not necessarily preserve geometric mean.
- Transform coding goal $\rightarrow$ reduce GM while AM stays fixed, because optimal **distortion** depends on GM.

—

4 Wavelet-Based Image Coding

Domanda 29

State the time-frequency uncertainty principle ($\Delta t \cdot \Delta f \geq 1/4\pi$), explain why it imposes a trade-off, and how wavelets address it with adaptive multiresolution (short windows at high frequencies, long at low frequencies).

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

- Short window $\rightarrow$ good time/space localization, poor frequency resolution.
- Long window $\rightarrow$ good frequency resolution, poor localization.
- Wavelets $\rightarrow$ multiresolution:
- high frequencies $\rightarrow$ short basis functions;
- low frequencies $\rightarrow$ long basis functions.
- Image match $\rightarrow$ edges need localization; smooth trends need coarse-scale representation.

Domanda 30

Compare STFT (rigid tiling) and **DWT** (adaptive tiling) of the time-frequency plane, linking them to the trends vs anomalies image model.

- **STFT** $\rightarrow$ fixed window size $\rightarrow$ rigid time-frequency tiling; same resolution for all frequencies.
- **DWT** $\rightarrow$ filter banks + downsampling $\rightarrow$ low-frequency approximations + high-frequency detail subbands.
- Repeated **DWT** $\rightarrow$ multiresolution representation.
- Image model:
- smooth trends $\rightarrow$ coarse low-frequency components;
- anomalies/edges $\rightarrow$ high-frequency localized details.

Domanda 31

(TC&JPEG) What is a primary advantage of the hierarchical (multiresolution) decomposition offered by the **wavelet** Transform in JPEG2000?

- **wavelet** decomposition $\rightarrow$ progressive and multiresolution coding.
- Low-resolution approximation decoded first $\rightarrow$ detail subbands refine quality and resolution.

Domanda 32

(TC&JPEG) Compare the block-based **DCT** approach used in JPEG with the **wavelet**-based decomposition used in JPEG2000, specifically in terms of how they handle the image signal and the resulting artifacts.

(TC&JPEG) Which of the following is a key reason why JPEG2000 is generally more efficient than JPEG?

- **JPEG:**
 - independent 8×8 block **DCT**;
 - simple and efficient;
 - low bitrate $\rightarrow$ **blocking artifacts**.
- **JPEG2000:**
 - **DWT** over large tiles / whole image;
 - multiresolution representation;
 - embedded bitplane coding and precise **rate** control;
 - low bitrate $\rightarrow$ ringing/blurring near edges, not blocking.
 - Efficiency reasons:
 - **wavelet** multiresolution decomposition;
 - embedded bitstream truncation;
 - independent codeblocks;
 - quality/resolution scalability;
 - no fixed 8×8 boundaries.

Domanda 33

Describe the JPEG2000 architecture (Tier 1: **DWT** 9/7 or 5/3 $\rightarrow$ fine **quantization** $\rightarrow$ arithmetic coding of codeblocks per bitplane; Tier 2: **EBCOT**). Where does the **lossy** operation actually happen?

- **JPEG2000** $\rightarrow$ **wavelet** still-image codec for scalability, **rate** control, ROI, **lossy/lossless**.
- Tier 1:

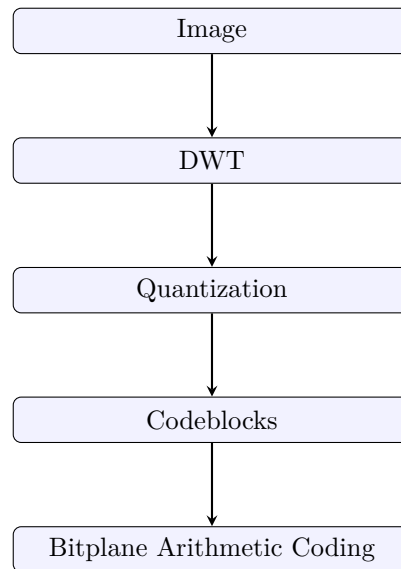


Figure 6: JPEG2000 Tier 1 Coding Chain

- 9/7 **DWT** → irreversible, **lossy**.
- 5/3 **DWT** → reversible, **lossless**.
- Coefficients → codeblocks → coded bitplane by bitplane.
- Tier 2 / **EBCOT** → layers, packets, progression orders, truncation points.
- **lossy** operations → **quantization** and embedded bitplane/pass truncation.

Domanda 34

Compare JPEG and JPEG2000 on channel-error robustness (codeblock independence, contained vs catastrophic propagation, resynchronization markers).

- **JPEG:**
 - sequential **entropy** coding;
 - bit error can desynchronize Huffman stream;
 - corruption propagates until restart/resynchronization marker.
- **JPEG2000:**
 - relatively independent codeblocks;
 - stream organized into packets/layers;
 - damage tends to remain local to codeblock/**subband**/quality layer.
- Result → JPEG2000 more robust to localized channel errors.

5 Learned Image Coding

Domanda 35

(TC&JPEG) Explain the fundamental shift in **rate-distortion** (R-D) optimization from classical codecs (e.g., JPEG) to neural compression methods.

- Classical codecs → optimize handcrafted tools:
- fixed transforms;
- prediction modes;
- block partitions;
- **quantization** steps.

$$J = D + \lambda R$$

- Neural compression → learns analysis transform, synthesis transform, **entropy** model.

$$\mathcal{L} = D(x, \hat{x}) + \lambda R(\hat{y})$$

- Main shift → data-learned transform instead of fixed standard tools.

Domanda 36

JPEG-AI: goals (beat **VVC-Intra** by about 50%), backbone (hierarchical VAE with hyperprior), dual-use human/machine support, and complexity profiles (Dec0/Dec1/Dec2, kMAC/px). Cite pros/cons vs classical codecs (R-D vs computational cost, determinism, hallucinations).

- **JPEG-AI** → learned still-image coding for human viewing and machine consumption.
- Goal → beat classical intra codecs, target about **50% gain vs VVC-Intra** in favorable settings; notes also report about **27% saving at 0.5 bpp**.
- Backbone:

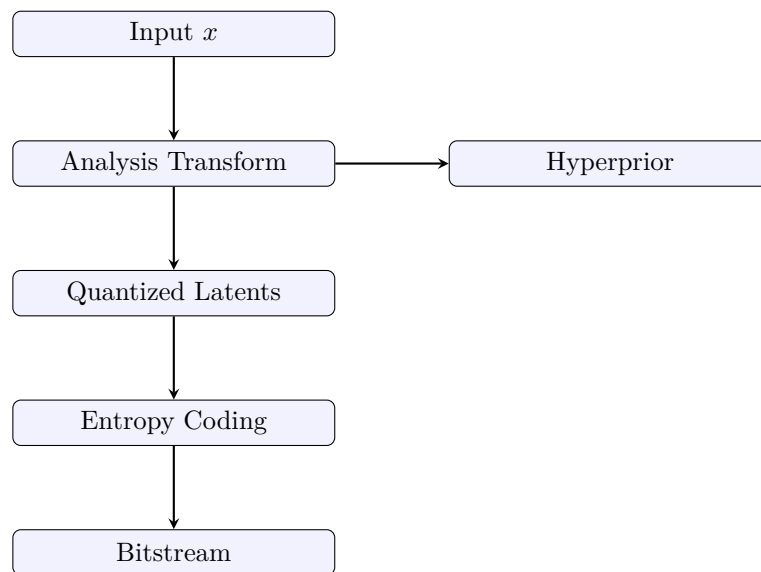


Figure 7: JPEG-AI Hierarchical VAE Backbone

- Hyperprior → side information about latent statistics (scales/probabilities); costs bits but improves **entropy** modeling.
- Backbone class → hierarchical **VAE** with hyperprior; nonlinear analysis/synthesis transforms learned end-to-end.
- Dual use → support both **human viewing** and **machine consumption**.
- Decoder profiles, complexity in **kMAC/px**:
- **Dec0** → about 8 kMAC/px;
- **Dec1** → about 23 kMAC/px;
- **Dec2** → about 214 kMAC/px.
- Pros:
- better R-D efficiency;
- nonlinear transforms learned from data;
- possible human/machine task support.
- Cons:
- high computational/energy cost;
- harder deterministic reproducibility;
- risk of plausible but inaccurate reconstructed details.

Domanda 37

(TC&JPEG) What is the primary purpose of adding Additive Uniform Noise during the training phase of a neural codec?

- **quantization** → non-differentiable, zero gradient almost everywhere.

- Training relaxation $\rightarrow$ additive uniform noise:

$$u \sim \mathcal{U}\left(-\frac{1}{2}, \frac{1}{2}\right)$$

- Effect $\rightarrow$ approximates rounding with continuous operation $\rightarrow$ gradient-based optimization possible.

Domanda 38

(TC&JPEG) Why do Convolutional Neural Networks (CNNs) outperform Multi-Layer Perceptrons (MLPs) when applied to image compression?

- MLPs $\rightarrow$ flatten images, ignore spatial locality, many parameters, weak image bias.
- CNNs $\rightarrow$ exploit local correlations, translation structure, shared filters.
- Result $\rightarrow$ more efficient image compression models.

—

6 Audio and Speech Coding

Domanda 39

Explain masking in the auditory system: define frequency masking and **temporal masking** (pre/post-masking) with their time-scale orders of magnitude.

- **Auditory masking** → strong sound makes weaker sound inaudible.
- Codec use → hide **quantization** noise below **masking threshold**.
- **Frequency masking** → **simultaneous masking**; strong component at f_0 raises threshold around nearby frequencies, mostly same critical band.
- **temporal masking**:
 - pre-masking → few ms before masker;
 - post-masking → up to about 100-200 ms after masker, depending on level/content.

Domanda 40

What is the critical band and how does it relate to the audibility condition of a set of sinusoids close in frequency?

- **Critical band** → frequency interval processed roughly by same cochlear auditory filter.
- Bandwidth increases with center frequency.
- Close sinusoids interact inside same band.
- Audibility condition:

$$\sum_i g_i^2 > S_0(f_n)$$

- g_i^2 → sinusoid powers.
- $S_0(f_n)$ → absolute hearing threshold near frequency f_n .

Domanda 41

(A&S Comp.) Explain the difference between Source-Based (Parametric) coding and Sink-Based (Perceptual) coding.

- **Source-based / parametric coding** → models signal production.
- Speech → source-filter model of vocal folds + vocal tract.
- Goal → intelligibility and low latency at very low bitrate.
- **Sink-based / perceptual coding** → models listener perception.
- Audio codecs → auditory masking + critical bands.

- Goal → transparent quality by removing inaudible information.

Domanda 42

(A&S Comp.) Describe the “Analysis-by-Synthesis” (AbS) loop used in **CELP** (Code Excited Linear Prediction) codecs and why it represents an improvement over simple **LPC-10**.

- **LPC-10** → simplified excitation: impulse train for voiced, noise for unvoiced → synthetic sound.
- **CELP** → **Analysis-by-Synthesis** closed-loop search:

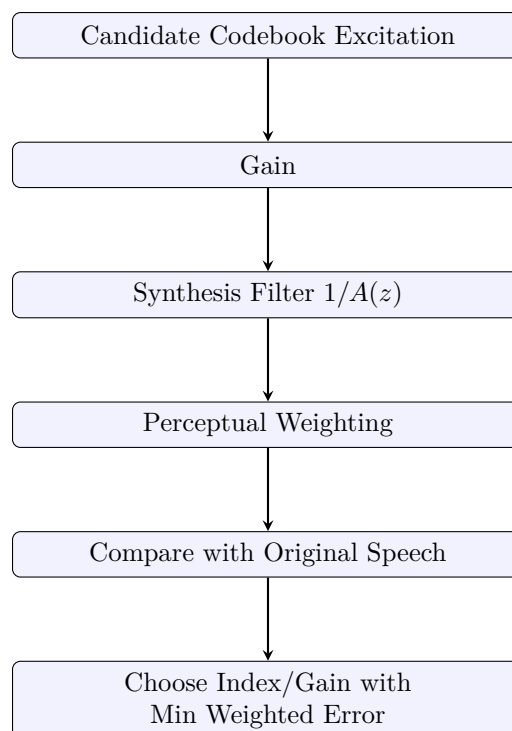


Figure 8: CELP Analysis-by-Synthesis Loop

- **bit allocation:**
- sensitive bands → more bits;
- masked bands → fewer bits.

Domanda 44

(A&S Comp.) Describe the principles of the LPC10 speech coding scheme.

- **LPC** → short speech frame modeled as excitation through all-pole vocal-tract filter.
- Encoder sends → filter parameters, gain, excitation information.

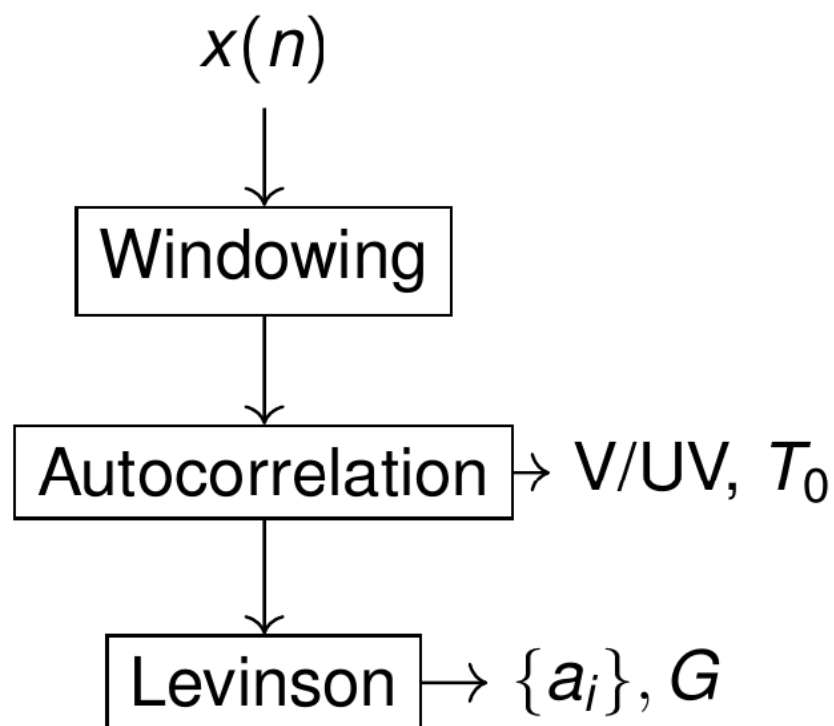
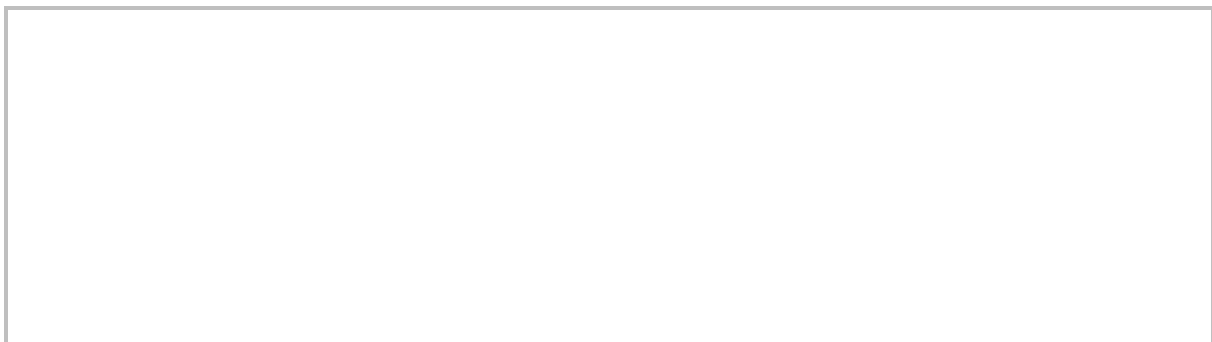


Figure 9: Linear predictive coding - analysis



- Analysis:

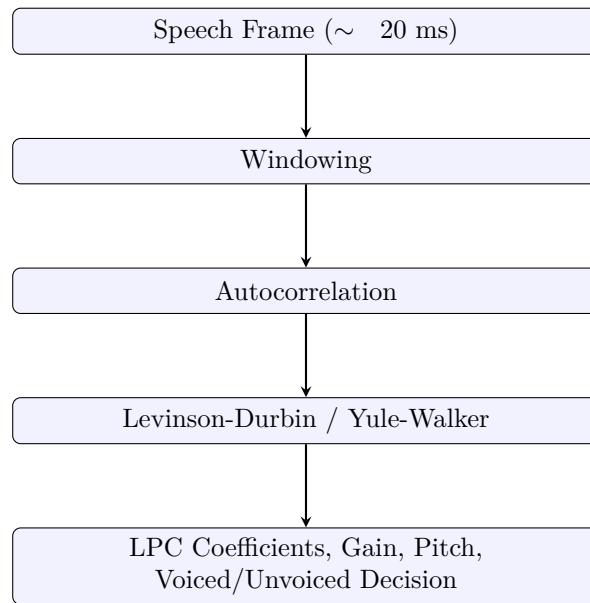


Figure 10: LPC Analysis Block Scheme

- Prediction model:

$$\hat{x}(n) = - \sum_{i=1}^P a_i x(n-i)$$

- Residual:

$$y(n) = x(n) - \hat{x}(n) = \sum_{i=0}^P a_i x(n-i), \quad a_0 = 1$$

- Voiced frames → **pitch**-period excitation.
- Unvoiced frames → noise-like excitation.
- **LPC-10** → low bitrate, less natural than **CELP** because excitation is rigid.

Domanda 45

(A&S Comp.) Draw the scheme and describe the operation of the functional blocks of an MP3 encoder.

- **MP3** → perceptual audio codec.

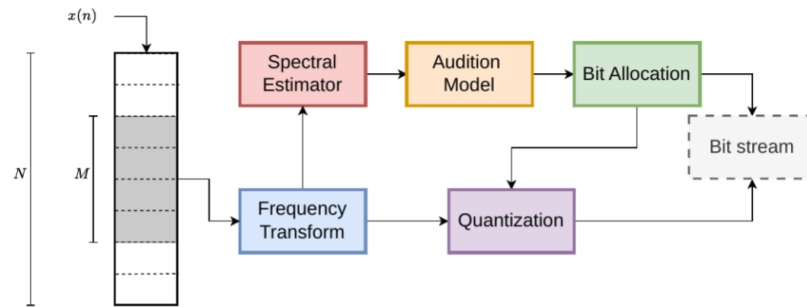


Figure 11: Pasted image 20260624185047

- Main encoder chain:

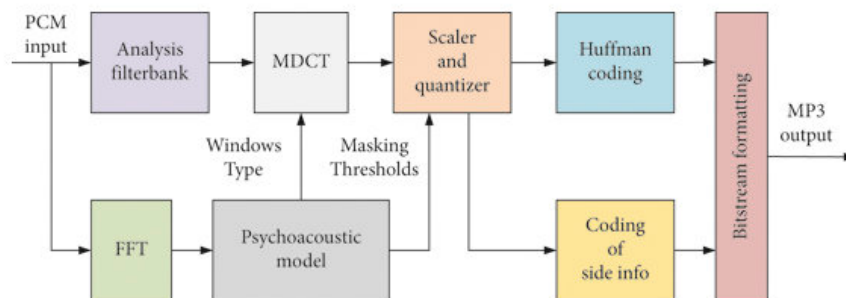


Figure 12: Pasted image 20260624213214

- Functional blocks:
- framing/windowing + transform/filterbank → convert time signal into spectral components;

- **psychoacoustic model** → estimate **masking threshold** from signal spectrum;
- **bit allocation** → spend more bits where ear is sensitive, fewer where noise is masked;
- **quantization** → **lossy** stage on spectral coefficients;
- **entropy coding** → final **lossless** compression stage.
- **psychoacoustic model** → encoder-side only; not needed at decoder.
- Decoder → **entropy** decoding, inverse **quantization**, inverse transform / synthesis filterbank.

Domanda 46

(A&S Comp.) Why are Line Spectrum Frequencies (LSF) preferred over direct **quantization** of **LPC** coefficients (a_i)?

- Direct **LPC** coefficient **quantization** → can create unstable synthesis filters.
- **LSF** → represent **LPC** filter through roots on unit circle.
- Stability condition easier to enforce by preserving interlacing/order:

$$0 < \omega_1^{(P)} < \omega_1^{(Q)} < \omega_2^{(P)} < \dots < \pi$$

- If **quantization** disturbs order → decoder can restore valid ordering.

Domanda 47

(A&S Comp.) Regarding the Opus audio codec, what is the primary technical advantage of its hybrid design?

- **Opus** combines:
- **SILK** → **LPC**-based, efficient for speech and low bitrates;
- **CELT** → **MDCT**-based, efficient for music/general audio and low delay.
- Advantage → seamless adaptation across speech/music, bitrate, latency; useful for WebRTC.

Domanda 48

(A&S Comp.) Which of the following best describes current research trends in the future of multimedia audio coding?

- Neural speech/audio codecs at very low bitrates.
- Perceptual and QoE-driven quality metrics.
- Spatial/immersive audio.
- Robust real-time coding under latency constraints.

- Quality assessment for generated/enhanced audio.

—

7 Quality Evaluation and Quantization

Domanda 49

Compare **MSE/PSNR**, **SSIM** and **LPIPS**: what each measures, pros/cons, and why two images with the same **MSE** can have very different perceived quality.

- **MSE** → average squared pixel error:

$$\text{MSE} = \frac{1}{NM} \sum_{i,j} (x_{ij} - \hat{x}_{ij})^2$$

- **PSNR** → logarithmic **MSE**:

$$\text{PSNR} = 10 \log_{10} \left(\frac{MAX^2}{\text{MSE}} \right)$$

- **MSE/PSNR** → simple and reproducible, but weak perceptually; ignore structure, masking, semantics.
- **SSIM** → compares local luminance, contrast, structure.
- **LPIPS** → compares deep features; better for perceptual/semantic distortions, but expensive and model-dependent.
- Same **MSE** can look different → error distribution matters: texture noise less visible; edge/structure error more annoying.

Domanda 50

(MMQuEv) Explain the difference between subjective and objective quality evaluation in multimedia systems and why both are necessary.

- **Subjective evaluation** → human observers **rate** perceived quality.
- Ground truth for perception.
- Slow, expensive, statistically variable.
- **Objective evaluation** → automatic signal/feature metric.
- Fast, repeatable, scalable.
- May correlate imperfectly with human perception.
- Need both → subjective validates perception; objective enables optimization and monitoring.

Domanda 51

(MMQuEv) What are the key stages involved in designing a subjective quality test according to standardized guidelines?

- Define:
- dataset/content selection;
- display/audio equipment;
- viewing/listening conditions;
- methodology: ACR, DSIS, pairwise comparison, etc.;
- participant screening;
- score processing/outlier handling.
- Control factors → viewing distance, brightness/contrast, room illumination, audio setup, test duration.

Domanda 52

(MMQuEv) Describe the main categories of objective quality metrics based on the availability of the original “reference” signal.

(MMQuEv) Which of the following best describes the “Full-Reference” (FR) objective quality assessment approach?

- **Full Reference (FR)** → original and degraded signals available; examples: **MSE**, **PSNR**, **SSIM**, **VMAF**.
- **Reduced Reference (RR)** → only compact original features available.
- **No Reference (NR)** → only degraded signal available; examples: **BRISQUE**, **NIQE**, **PIQE**.
- FR luminance **PSNR**:

$$\text{MSE}_Y(t) = \frac{1}{NM} \sum_{n,m} [I(n, m, 1, t) - \hat{I}(n, m, 1, t)]^2$$

$$\text{PSNR}_Y(t) = 10 \log_{10} \left(\frac{255^2}{\text{MSE}_Y(t)} \right)$$

Domanda 53

(MMQuEv) In the context of subjective testing, what is the main purpose of a “screening” phase for participants?

- **Screening** → checks participants can correctly perceive stimuli.
- Removes invalid observers: uncorrected visual issues, color-vision problems, inconsistent scoring behavior.

Domanda 54

(MMQuEv) Why is statistical analysis a critical component of subjective quality evaluation?

- Human scores vary → need reliability measures.
- Statistical analysis → **MOS**, variance, confidence intervals, outlier detection.
- **MOS**:

$$\text{MOS} = \frac{1}{N} \sum_{i=1}^N x_i$$

- Standard error:

$$SE = \frac{s}{\sqrt{N}}$$

Domanda 55

(S&P Qnt) Explain the difference between a “mid-tread” and a “mid-rise” **quantizer** in the context of **uniform quantization** for signed data.

- **Mid-tread** → zero is reconstruction level; small values around zero map to zero.
- **Mid-rise** → zero is decision threshold; values around zero map to positive/negative non-zero levels.
- Residuals prefer mid-tread → many small values become exactly zero:

$$Q(x) = \Delta \cdot \text{round}\left(\frac{x}{\Delta}\right)$$

Domanda 56

(S&P Qnt) Define the concept of a “**deadzone**” in a **quantizer** and explain why it is frequently employed in **lossy** compression systems.

- **deadzone quantizer** → enlarged interval mapped to zero:

$$Q(x) = 0 \quad \text{for small } |x|$$

- Useful in **lossy** compression → prediction/transform residuals often have many small coefficients.
- Consequence → more zeros → better **entropy** coding.

Domanda 57

(S&P Qnt) Why is scalar **quantization** alone often considered insufficient for effective compression of non-sparse data?

- Scalar **quantization** → one sample at a time.
- Non-sparse signal → many significant samples remain → **entropy** coding **HAS** little to exploit.
- Better compression needs:
 - prediction → reduce residual variance;
 - transform coding → compact energy;
 - vector/block coding → exploit dependencies;

- **entropy** coding $\rightarrow$ exploit non-uniform probabilities.

Domanda 58

(S&P Qnt) What is the condition for a **predictive quantization** system to be effective, and how is the “coding gain” defined?

- Effective condition:

$$\sigma_y^2 < \sigma_x^2$$

with:

$$y(n) = x(n) - v(n)$$

- Prediction gain:

$$G_P = 10 \log_{10} \left(\frac{\sigma_x^2}{\sigma_y^2} \right)$$

- Positive when predictor reduces variance.

Domanda 59

(S&P Qnt) Draw the scheme of a linear **predictive quantization** scheme, and motivate the structure, with particular attention to the use of a decoding loop at the encoder side.

- Predictor must use reconstructed past samples on encoder and decoder:

$$v(n) = \mathcal{P}(\hat{x}(n-1), \hat{x}(n-2), \dots)$$

$$y(n) = x(n) - v(n)$$

$$\hat{x}(n) = \hat{y}(n) + v(n)$$

- Encoder includes same reconstruction loop as decoder $\rightarrow$ prevents drift.

Domanda 60

(S&P Qnt) What is the primary purpose of the predictor in a **predictive quantization** system?

(S&P Qnt) In a **predictive quantization** system, if the prediction $v(n)$ is nearly equal to $x(n)$, what happens to the variance of the signal $y(n)$ being sent to the **quantizer**?

- Predictor $\rightarrow$ exploits correlation among neighboring samples.

- If $v(n) \approx x(n)$:

$$y(n) = x(n) - v(n) \approx 0$$

$$\sigma_y^2 \ll \sigma_x^2$$

- **quantizer** encodes lower-energy residual.

Domanda 61

(S&P Qnt) When selecting a linear predictor of order P for a random process, how does the prediction error variance typically behave as the order increases?

- Linear predictor:

$$v(n) = - \sum_{i=1}^P a_i x(n-i)$$

$$y(n) = \sum_{i=0}^P a_i x(n-i), \quad a_0 = 1$$

- Increasing P cannot worsen optimal error variance in theory $\rightarrow$ old solution still available.
- Practice $\rightarrow$ larger P increases complexity, side information, estimation error; gains eventually negligible.

Domanda 62

Explain the screening effect: why does the prediction gain saturate as the linear predictor order P increases?

- Increasing $P \rightarrow$ optimal prediction error variance cannot increase in theory.
- Gain usually saturates due to **screening effect**.
- Nearby samples already explain most correlation with current sample.
- Farther samples add little independent information because correlation is mostly mediated by nearer samples.
- Result $\rightarrow$ diminishing returns plus higher complexity/estimation sensitivity.

Domanda 63

(S&P Qnt) In high-resolution **uniform quantization**, what is the approximate relationship between the SNR and the bit **rate** (R)?

- High-resolution **uniform quantization**:

$$D \approx \frac{A^2}{12} 2^{-2R}$$

$$\text{SNR} = 10 \log_{10} \left(\frac{\sigma_X^2}{D} \right) = 10 \log_{10} \left(\frac{\sigma_X^2}{\frac{A^2}{12} 2^{-2R}} \right)$$

- With $\gamma^2 = \frac{A^2}{4\sigma_X^2}$:

$$\text{SNR} \approx 6.02R - 10 \log_{10} \left(\frac{\gamma^2}{3} \right)$$

- Rule $\rightarrow$ each extra bit/sample gives about 6 dB SNR improvement.

—

8 Adaptive Streaming

Domanda 64

Compare QoS and QoE: what they measure and how they correlate. Which network factors impact streaming QoE?

- **QoS** → technical service/network metrics: throughput, latency, jitter, packet loss.
- **QoE** → user-perceived service quality: media quality, startup delay, stalls, quality switches, device, context, expectations.
- Correlation → not linear.
- Packet loss may be hidden by buffer.
- Short throughput drop may cause rebuffering → large QoE loss.

Domanda 65

(AdapStrm) Describe the fundamental architectural differences between a “Push-based” streaming system (e.g., RTP/UDP) and a “Pull-based” system (e.g., **DASH**).

- **Push-based streaming** (RTP/UDP/WebRTC-style):
 - server sends packets according to timing;
 - low latency;
 - needs real-time transport/control;
 - typical use → conferencing/live interactive media.
- **Pull-based streaming** (**DASH/HLS** over HTTP):
 - client requests segments;
 - works with HTTP/CDNs/caches;
 - client selects bitrate from throughput/buffer;
 - higher delay, scalable/robust for VoD/streaming.

Domanda 66

(AdapStrm) Analyze the role of the client-side buffer in the context of stability and Quality of Experience (QoE).

- Buffer → stores downloaded media in seconds.
- Function → absorbs throughput variation and jitter.
- Larger buffer → more stability, but higher startup delay/latency.
- QoE depends on startup time, stalls, stall duration, segment quality, quality switches.

Domanda 67

(AdapStrm) Define “Switching Penalty” and discuss its impact on perceived video quality.

- **Switching penalty** → QoE loss from visible quality changes between segments.

$$J(n) = \lambda_1 K_n - \lambda_2 |K_n - K_{n-1}| - \phi(\Delta_n)$$

- $K_n \rightarrow$ segment quality.
- $|K_n - K_{n-1}| \rightarrow$ switching penalty.
- $\phi(\Delta_n) \rightarrow$ rebuffering penalty.
- Frequent oscillations can be worse than stable slightly lower quality.

Domanda 68

(AdapStrm) Explain the evolution of the playout buffer level $B(t)$ using a mathematical model. In your explanation, describe the dynamics of the “playback” (draining) phase and the “rebufferization” (stalling) phase.

- Buffer level in playback seconds:

$$B(t) = L \cdot T_s$$

- $L \rightarrow$ stored segments; $T_s \rightarrow$ segment duration.
- Coding **rate** R_c , throughput S :

$$\frac{dB}{dt} = \begin{cases} \frac{S}{R_c} - 1 & \text{during playback} \\ \frac{S}{R_c} & \text{during rebuffering} \end{cases}$$

- Playback → buffer receives data but drains at one playback second per real second.
- Rebuffering → playout stops; no output drain.
- Stable playback average:

$$S > R_c$$

Domanda 69

(AdapStrm) What is the primary motivation for using HTTP-based protocols for video streaming?

- HTTP streaming → stateless, CDN-friendly, firewall-friendly, easy to deploy at scale.
- **DASH/HLS** → use web infrastructure; client adapts representation quality.

Domanda 70

(AdapStrm) What is the consequence of an **ABR** algorithm that systematically overestimates the available bandwidth?

- **ABR** overestimates throughput → requests too high bitrate.
- Segment download time grows → buffer empties → rebuffering/stalling.

Domanda 71

(AdapStrm) Which metric is a direct indicator of streaming QoE from the end-user's perspective?

- Direct QoE indicators → startup delay, rebuffering duration/frequency, quality level, quality switches.
- User perspective → uninterrupted stable playback often more important than low-level network metrics.

Domanda 72

(AdapStrm) At what stage in the session lifecycle does a **DASH** client process the Media Presentation Description (MPD)?

- Client processes **MPD** at session start, before media segment download.
- MPD lists periods, adaptation sets, representations, codecs, bitrates, resolutions, segment URLs.

Domanda 73

Explain how an **ABR** (Adaptive Bitrate) algorithm works: **rate-based** vs **buffer-based** logic, and the risk of overestimating available bandwidth.

- **ABR** → client-side rule selecting bitrate representation for next segment.
- Goal → maximize QoE: quality + stability, avoid stalls.

- **rate-based:**
 - estimate recent throughput;
 - choose representation below available bandwidth, often with safety margin.
 - **buffer-based:**
 - low buffer → conservative bitrate;
 - high buffer → higher quality.
 - Overestimated bandwidth → segments too large → long downloads → buffer drain → stalls.
-

9 Motion Estimation and Video Coding

Domanda 74

motion estimation: give the principles of the block matching approach. Give at least one cost function. [Bonus]. Discuss the regularization issue.

- **Block matching** → split frame into blocks $B_{p,q}$; search similar block in reference frame window.
- Best displacement:

$$(\hat{i}, \hat{j}) = \arg \min_{(i,j) \in W} J(i, j)$$

- SSD:

$$J_{\text{SSD}}(i, j) = \sum_{(n,m) \in B_{p,q}} [f(n, m, k) - f(n - i, m - j, h)]^2$$

- SAD:

$$J_{\text{SAD}}(i, j) = \sum_{(n,m) \in B_{p,q}} |f(n, m, k) - f(n - i, m - j, h)|$$

- Regularized:

$$J_{\text{REG}}(i, j) = \|\vec{f}_k(B_{p,q}) - \vec{f}_h(B_{p-i,q-j})\|_p^p + \lambda R(i, j)$$

- Regularization → penalizes expensive/irregular motion vectors; balances prediction quality with motion-vector coding **rate**.

Domanda 75

Discuss the advantages and the disadvantages of Intra, Predictive, and Bidirectional images in a GOP for video coding.

- **I images:**
 - intra only;
 - advantages → random access, error reset, no **motion estimation**;
 - disadvantages → high **rate**, lower compression.
- **P images:**
 - predicted from previous anchor frames;
 - advantages → good compression, moderate delay;
 - disadvantages → ME/MC complexity, error propagation.
- **B images:**

- predicted from past and future frames;
- advantages → best compression;
- disadvantages → highest complexity, structural delay, not ideal for low latency.
- GOP structure controls compression efficiency, delay, random access interval, error propagation.

Domanda 76

(ME) Describe the difference between “motion field” and “optical flow”.

- **Motion field** → projection of actual 3D scene motion onto image plane.
- **Optical flow** → apparent motion of brightness patterns.
- They often coincide, but can differ due to illumination changes, occlusions, texture ambiguities.

Domanda 77

(ME) Explain the Horn and Schunck algorithm’s core principle for dense optical flow estimation.

- Optical flow constraint:

$$uf_x + vf_y + f_t = 0$$

- Horn-Schunck objective:

$$J = \iint_{\mathcal{R}} (uf_x + vf_y + f_t)^2 dx dy + \lambda \iint_{\mathcal{R}} (\|\nabla u\|^2 + \|\nabla v\|^2) dx dy$$

- First term → brightness consistency.
- Second term → smooth dense flow field.

Domanda 78

(ME) Discuss the **rate-distortion** trade-off when selecting block sizes in **motion estimation**.

$$J(v) = d(B_k^{(p)}, B_h^{(p+v)}) + \lambda_{ME} R(v)$$

- Large blocks:
- fewer motion vectors;
- lower side information and complexity;
- worse fit near object boundaries.

- Small blocks:
- better prediction and lower **distortion**;
- more motion vectors and partition bits;
- higher complexity.

Domanda 79

(ME) Which of the following is a disadvantage of using the Sum of Squared Differences (SSD) as a matching criterion?

- SSD squares errors $\rightarrow$ outliers dominate cost.
- Illumination changes, noise, occlusions $\rightarrow$ irregular motion vectors.
- SAD $\rightarrow$ often more robust and cheaper.

Domanda 80

(ME) What is the main benefit of the Hexagon Search strategy compared to Full Search?

- Full search $\rightarrow$ tests all candidates; optimal within window, expensive.
- Hexagon search $\rightarrow$ tests small hexagonal pattern iteratively; far fewer evaluations.
- Benefit $\rightarrow$ faster search.
- Cost $\rightarrow$ no guaranteed global optimum.

Domanda 81

(ME) What does an affine motion model allow that a pure translational model does not?

- Translation $\rightarrow$ constant displacement only.
- Affine motion $\rightarrow$ rotation, zoom, shear:

$$\vec{v}(p) = \vec{b} + Bp = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + \begin{bmatrix} b_3 & b_4 \\ b_5 & b_6 \end{bmatrix} p$$

- If $B = 0 \rightarrow$ pure translation / block matching.

Domanda 82

Draw the block diagram of a hybrid video encoder (**motion estimation**/compensation + **DCT** + **quantization** + **entropy** coding + reconstruction loop with frame buffer). Explain why the encoder contains an internal decoder.

- Hybrid video encoder $\rightarrow$ codes prediction residual, usually not whole frame.

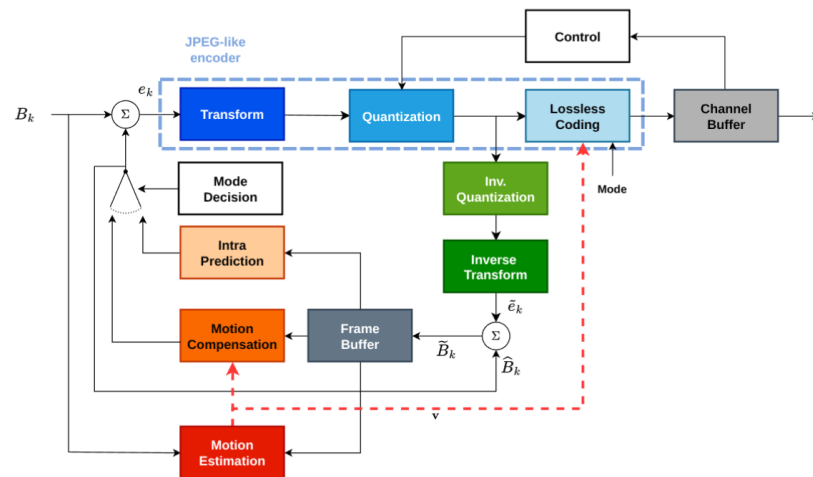


Figure 13: Hybrid video encoder

- Mode decision $\rightarrow$ selects intra/inter predictor.
- Transform + **quantization** $\rightarrow$ residual coding.
- lossless** coding $\rightarrow$ bitstream.
- Inverse **quantization** + inverse transform $\rightarrow$ reconstructed residual.
- Reconstructed block $\rightarrow$ frame buffer for later **motion compensation**.
- Internal decoder needed $\rightarrow$ future predictions must use same reconstructed frames as decoder; otherwise drift propagates.

Domanda 83

(VCP) Why is the temporal prediction error usually more efficient to encode than the original video signal?

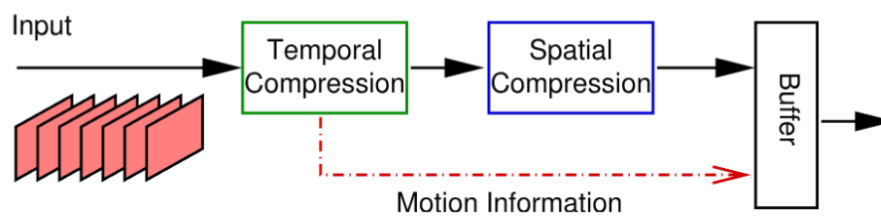


Figure 14: Video coding principles

- Video frames $\rightarrow$ highly correlated.
- Motion-compensated prediction estimates current block from reference frames.
- Encode only residual:

$$e = x - x_p$$

- Residual $\rightarrow$ lower energy and more sparse than original frame.
- Consequence $\rightarrow$ transform **quantization** and **entropy** coding more efficient.

Domanda 84

(VCP) Describe the function of the “Mode Selection” step in a hybrid video encoder.

- Mode selection $\rightarrow$ choose coding mode minimizing R-D cost:

$$J = D + \lambda R$$

- For blocks:

$$D = \sum_{k=1}^K D_k(i_k, Q), \quad R = \sum_{k=1}^K R_k(i_k, Q)$$

$$J(\vec{i}, Q, \lambda) = D(\vec{i}, Q) + \lambda R(\vec{i}, Q)$$

- Block-wise suboptimal minimization:

$$i_k^* = \arg \min_{i_k} J_k(i_k, Q, \lambda)$$

- Empirical choices:

$$\text{MPEG-2: } \lambda = aQ^2 + b$$

$$\text{H.264: } \lambda = c \cdot 2^{dQ+e}, \quad \lambda_{ME} = \sqrt{\lambda}$$

Domanda 85

(VCP) How does the “Channel Buffer” controller manage the trade-off between target **rate** and video quality?

- Channel buffer → controls **quantization** step to meet target **rate**.
- Buffer occupancy high → increase **quantization** step → reduce **rate**, lower quality.
- Buffer occupancy low → decrease **quantization** step → improve quality, higher **rate**.
- Goal → avoid overflow/underflow while keeping quality high.

Domanda 86

(VCP) What is the primary role of an “I-frame” in a GOP structure?

- **I-frame** → intra-coded and independently decodable.
- Roles → random access, GOP start, temporal error-propagation limit.

Domanda 87

(VCP) In the context of **motion vector** coding, why is a Median Predictor (MVP) used?

- Neighboring motion vectors → correlated.
- Median predictor estimates current vector from adjacent blocks.
- Encoder sends only difference:

$$\text{MVD} = \text{MV} - \text{MVP}$$

- MVD usually small → cheaper to encode.

Domanda 88

(VCP) What happens in the decoder when it receives an “Inter-coded” block?

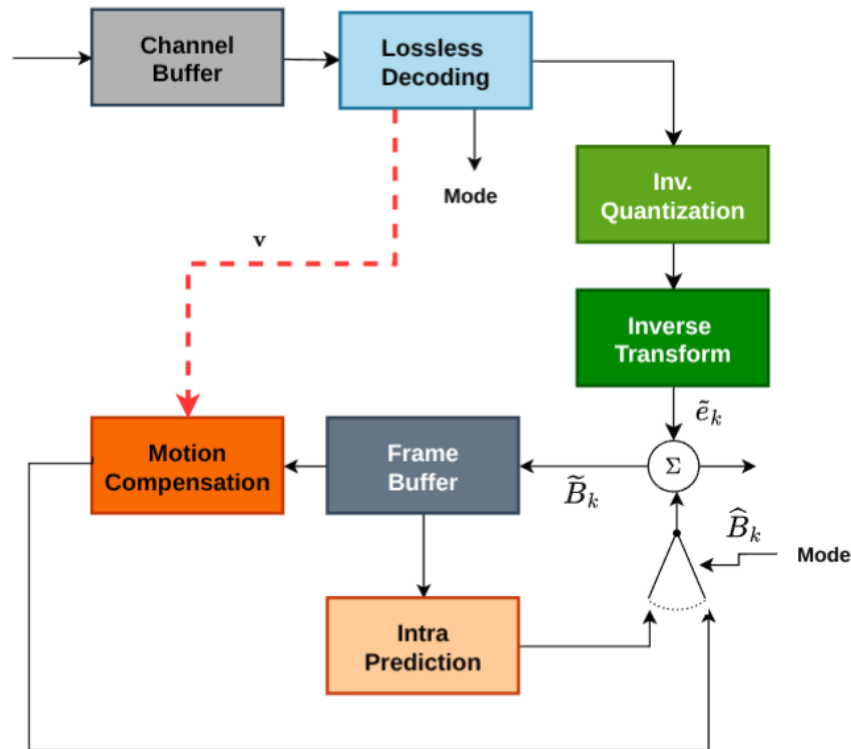


Figure 15: Hybrid video decoder

- Decoder reads side information $\rightarrow$ mode, reference index, **motion vector**.
- **entropy** decoding $\rightarrow$ residual and side information.
- Inverse **quantization** + inverse transform $\rightarrow$ reconstructed residual.
- **motion compensation** / **intra prediction** $\rightarrow$ predictor.
- Reconstruction:

$$\hat{x} = \text{prediction} + \hat{e}$$

- Output stored in frame buffer.

10 Modern Video Coding

Domanda 89

(ModernVC) What is the specific scope of video compression standards like H.266/**VVC**?

- Standards define:
- bitstream syntax;
- normative decoder behavior.
- They do not prescribe full encoder implementation.
- **VVC** → higher compression efficiency than **HEVC**, especially high-resolution/HDR/360/immersive video; cost → much higher complexity.

Domanda 90

(ModernVC) Explain the advantage of the “Coding Tree Unit” (CTU) structure introduced in **HEVC/VVC**.

- **CTU** → replaces fixed macroblocks with flexible recursive partitioning.
- Large homogeneous regions → large blocks.
- Detailed regions → smaller blocks.
- Result → improved R-D efficiency.
- **HEVC** → quadtree partitioning.
- **VVC** → multi-type trees, including binary and ternary splits.

Domanda 91

(ModernVC) What are the roles of VCL and NAL in modern video standards?

- **VCL (Video Coding Layer)** → compressed video data: slices, prediction, residuals, coding syntax.
- **NAL (Network Abstraction Layer)** → wraps VCL/non-VCL data into NAL units for transport/storage.
- Purpose → separates codec syntax from packetization.
- Non-VCL examples → SPS, PPS, SEI.

Domanda 92

(ModernVC) What is the main purpose of the **CABAC entropy** coder in modern standards?

- **CABAC** = Context-Adaptive Binary Arithmetic Coding.
- Workflow → binarize syntax elements, adapt probabilities to local context, arithmetic-code bins.
- Purpose → **lossless entropy** coding with better compression than VLC/**CAVLC**.
- Cost → higher complexity.

Domanda 93

(ModernVC) Why are “Tiles” considered “hardware-friendly” in **VVC** and **HEVC**?

- Tiles → independently decodable rectangular frame regions.
- Restrict dependencies across tile boundaries.
- Enable parallel encoding/decoding on CPU/GPU cores.
- Result → lower synchronization cost.

Domanda 94

(ModernVC) What is the function of an “In-loop filter” like the Adaptive loop filter (**ALF**)?

- In-loop filters → applied inside reconstruction loop before frames are stored as references.
- Purpose → reduce artifacts before **motion compensation** reuses frames.
- Evolution:
- H.264 → deblocking;
- **HEVC** → **SAO**;
- **VVC** → **ALF** and other tools.
- **ALF** → adaptive filtering of reconstructed samples; reduces ringing/residual **distortion**.

Domanda 95

Describe the intra-coding modes in H.264. [Optional] Discuss also the Intra modes in H.265.

- **intra prediction** → exploits already reconstructed neighboring samples in same frame.
- H.264/AVC:
- 4×4 blocks → 9 modes: 8 directional + DC;
- 16×16 **luma** blocks → 4 modes.
- H.265/**HEVC** → 35 modes: 33 directional + DC + Planar.
- H.266/**VVC** → 65 directional modes + wide-angle variants for non-square blocks.

- MPM lists → predict likely mode from neighboring blocks → reduce signaling cost.

Domanda 96

Describe the principle of the deblocking in-loop **filter** of H.264.

- Low bitrate block transform + **quantization** → visible block-boundary discontinuities.
- H.264 → normative in-loop **deblocking filter** on edges between 4×4 blocks.
- Filtering strength depends on:
 - coding mode;
 - motion vectors;
 - reference frames;
 - **quantization** parameter;
 - local boundary conditions.
- In-loop reason → filtered reconstructed frames become future references → avoids propagating blocking artifacts.